

Microstructure-Based Fair Value and Intraday Statistical Arbitrage in XLK

Final report: negative tradable result under incomplete-basket implementation

The active sparse microprice-signal strategy is positive in the January-February selection period but fails in the March out-of-sample period once executable midpoint P&L; and bid-ask costs are imposed. The no-trade benchmark is the best honest March decision.

Selected Holdings

symbol	official_weight_pct	basket_weight
NVDA	15.4537	0.2886
AAPL	12.6167	0.2356
MSFT	9.7708	0.1825
AVGO	5.9634	0.1114
AMD	3.4364	0.0642
CSCO	2.6566	0.0496
ORCL	2.3083	0.0431
CRM	1.3379	0.0250

Minute Data Diagnostics

symbol	minutes	median_spread_bps	trade_count
AAPL	20861	2.97	35436759
AMD	19553	9.33	24537293
AVGO	18860	10.19	22154477
CRM	18530	12.50	8300230
CSCO	22670	2.56	11425050
MSFT	20797	5.15	41606645
NVDA	22693	1.19	138130510
ORCL	18190	7.63	18803754
XLK	23748	0.73	15447039

Sparse Candidate Ranking

subset	train_adf_p	train_half_life_minutes	train_avg_oneway_cost_bps
MSFT NVDA ORCL CRM AMD	0.0149	187.7701	4.4528
NVDA ORCL CRM AMD	0.0221	225.3119	4.1037
NVDA AVGO ORCL CRM AMD	0.0295	196.3795	4.6313
MSFT NVDA AVGO ORCL AMD	0.0471	198.3262	4.5497
NVDA AVGO ORCL AMD	0.0565	233.8761	4.1845

Final Backtest

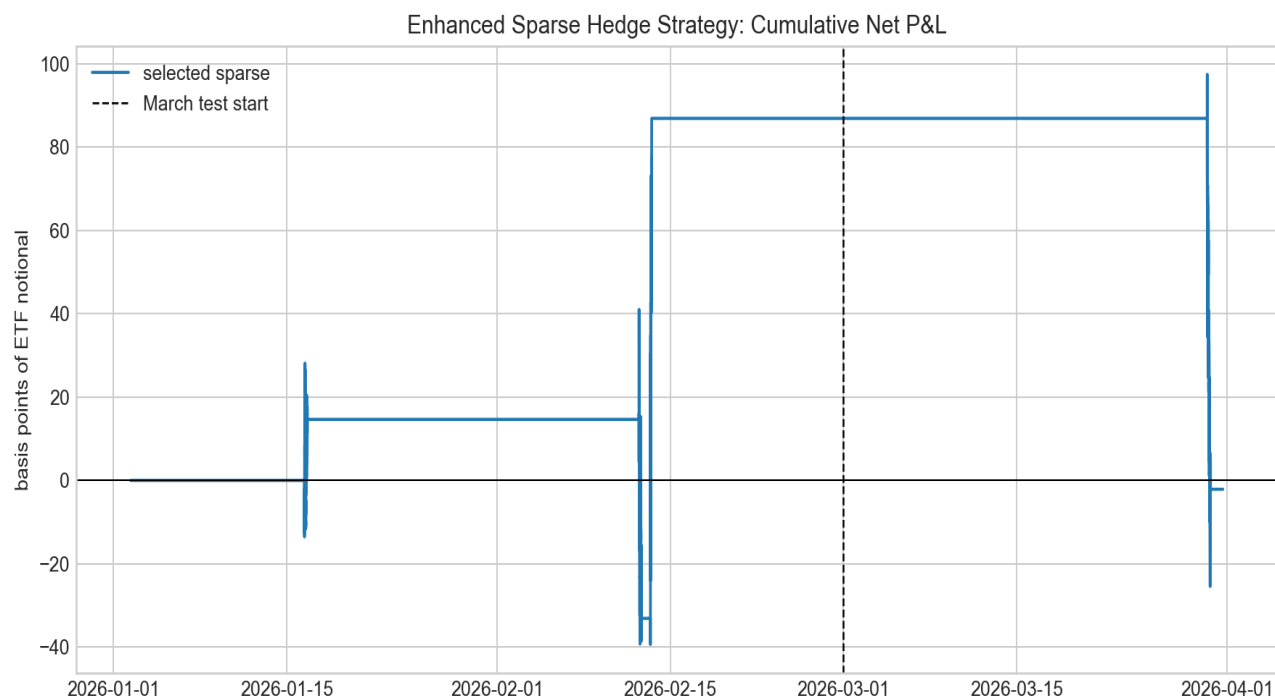
strategy	sample	trades	gross_bps	cost_bps	net_bps
sparse_e3.5_x1_plain	test	2.000	-82.083	6.950	-89.033
sparse_e3.5_x1_plain	train	6.000	117.726	30.797	86.930
literature_no_trade	test	0.000	0.000	0.000	0.000
literature_no_trade	train	0.000	0.000	0.000	0.000

Model Comparison

model	train_net_bps	march_net_bps	full_net_bps
baseline_original_best			-579.651
v2_best_by_train_overall	-48.200	-47.636	-95.835
v2_best_march_diagnostic	-118.168	54.077	-64.091
hybrid_sparse_micro_signal_mid_exec	86.930	-89.033	-2.103
literature_no_trade	0.000	0.000	0.000

Selection Audit

model	train_net_bps	march_net_bps	bias_flag
v2_best_by_train_overall	-48.200	-47.636	No test leakage.
v2_best_march_diagnostic	-118.168	54.077	Selection-on-test/look-ahead bias.
hybrid_sparse_micro_signal_mid_exec	86.930	-89.033	No test leakage in selection.
literature_no_trade	0.000	0.000	None.



Conclusion

The proposed ETF-basket arbitrage fails under the available eight-name proxy basket. Microprice is useful as a fair-value signal and diagnostic input, but the executable out-of-sample strategy does not beat no-trade after costs.